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United States Patent Application Publication	20250283400
Kind Code	A1
Publication Date	September 11, 2025
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MULTI-STAGE, LIMITED ENTRY DOWNHOLE GAS SEPARATOR

Abstract

A gas separator for use in a wellbore. The separator has an annular, staged separator system, whereby a liquid-gas mixture is separated into its liquid and gas components. Gas components are allowed to escape up the annulus around the separator, while liquid components are captured on each stage and removed through an inner tube. The inner tube has limited entry ports. These ports may be sized such that ports at the top of the string have a smaller, more limited entry than those toward the bottom.

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Family ID: **1000008621108**

Appl. No.: **19/217363**

Filed: **May 23, 2025**

Related U.S. Application Data

parent US continuation 17473410 20210913 parent-grant-document US 12331628 child US 19217363
parent US continuation 16815913 20200311 parent-grant-document US 11131180 child US 17473410
us-provisional-application US 62816516 20190311

Publication Classification

Int. Cl.: **E21B43/38** (20060101); **E21B17/18** (20060101); **E21B43/12** (20060101)

U.S. Cl.:

CPC **E21B43/38** (20130101); **E21B17/18** (20130101); **E21B43/122** (20130101);

Background/Summary

BACKGROUND

[0001] When pumping from a hydrocarbon producing well containing gas and liquid it is known to be desirable to separate the gas from the liquid in order for the pump to operate effectively. Known gas separators have various deficiencies such that gas interference, resultant gas-locking, and potential resultant damages to downhole pumping equipment, as well as downtime and deferred production is an ongoing problem.

[0002] Some examples of gas separators are described in U.S. Pat. No. 6,932,160 by Murray et al, U.S. Pat. No. 7,055,595 by Mack et al, U.S. Pat. No. 4,676,308 by Chow et al, and U.S. Pat. No. 2,883,940 by Gibson et al. Known gas separator devices can typically have limited effectiveness while occupying large amounts of space within the interior diameter of the well casing such that insertion and removal from the well casing may be awkward and difficult, and/or limited access is provided for other downhole tools if desired.

SUMMARY

[0003] The present invention is directed to a gas separator assembly for use downhole in a wellbore. The assembly is configured for placement in an outer casing. The assembly comprises an elongate outer tube, an elongate inner tube, and a pump. The elongate outer tube is configured to be received within the outer casing. An outer annular region is defined between the outer tube and the outer casing. The elongate inner tube disposed within the outer tube and defines an inner passage within the inner tube. An inner annular region is defined between the inner tube and the outer tube. The pump is in communication with the inner passage. A plurality of separator stages are defined along a length of the separator assembly, in series, and each separator stage is defined by at least one limited entry port which communicates fluid from the inner annular region into the inner tube.

[0004] In another aspect, the present invention is directed to a separator assembly. The assembly comprises an elongate first tube, an elongate second tube, and a plurality of isolator segments. The first tube is in communication with a pump. The second tube is disposed about the first tube along its length to form a first annular region therebetween. The isolator segments are disposed within the first annular region, each segment separating the first annular region into adjacent separator stages. A plurality of intake openings are formed in the outer tube to interconnect the first annular region of each separator stage with a region external to the outer tube. At least one limited access port is formed in the inner tube to interconnect the inner tube with the first annular region of each separator stage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a schematic representation of a gas separator assembly having an upper discharge point with annular separation chamber and a series of lower entry points for fluid intake and production.

[0006] FIG. **2** is schematic representation of a gas separator assembly having an upper discharge point with both annular and a series of internal gas separation chambers for fluid intake and production.

[0007] FIG. **3** is a schematic representation of a gas separator assembly having a multi-stage section in series with a diversion or packer type separator.

[0008] FIG. **4** is a schematic representation of a gas separator assembly comprised of a series of stacked, individual separation stages having no diverter separator attached at its bottom.
[0009] FIG. **5** is a sectional side view of one section of the gas separator assembly of FIGS. **3** and **4**.
[0010] FIG. **6** is a sectional side view of one section of a gas separator assembly utilizing a modified outer tube for annular velocity disruption.
[0011] FIG. **7** is a sectional side view of one section of a gas separator, as in FIG. **6**, but with a greater number of openings for annular velocity disruption.

DETAILED DESCRIPTION

[0012] Referring to the accompanying figures, there is illustrated a gas separator assembly generally indicated by reference numeral **10**. The assembly **10** is particularly suited for use in a downhole wellbore for separation of gas and liquids from a multi-phase fluid. The downhole wellbore typically comprises an outer casing **12** extending longitudinally along the wellbore between a wellhead at the surface of the ground and a hydrocarbon formation in the ground. The gas separator assembly **10** is generally used together with a pump **14** for pumping fluids up the wellbore for collection at the surface. The pump **14** may be mounted at the bottom end of a tubing string **16** received longitudinally within the outer casing **12** with the gas separator assembly **10** in turn being supported on the bottom of the tubing string **16** below the pump **14**.
[0013] Although various embodiments of the gas separator assembly are illustrated in the accompanying figures, the features in common with the first three embodiments will first be described.
[0014] In each instance the assembly **10** includes an outer tube **18** which is adapted to be received within the outer casing **12** so as to extend in a longitudinal direction along a length of the outer casing. The outer tube **18** effectively defines an outer annular region **20** between the outer tube and the outer casing **12**. The outer tube **18** can be connected in series with the tubing string **16** thereabove at a location below the pump **14**. The outer tube **18** typically extends the length of a multistage separator section **22** of the assembly **10** as described in further detail below.
[0015] The assembly further includes an intake section **24** connected in series with the outer tube **18** at a location therebelow at the bottom end of the assembly **10**. The intake section includes an intake opening **26** at the bottom end thereof which is in communication with the remaining outer casing **12** extending below the assembly **10** which in turn communicates with a surrounding hydrocarbon formation.
[0016] An isolation plug **28** is provided for spanning the outer annular region **20** between the intake section **24** in the surrounding outer casing **12** for blocking direct communication of fluids from the portion of the outer casing below the isolation plug to the outer annular region **20** above the isolation plug.
[0017] An inner tube **30** is supported within the outer tube **18** at a location above the intake section **24** to define a separate, inner passage therein. The top end of the inner tube **30** is connected to the pump **14** such that fluids within the inner tube **30** are drawn upwardly by the pump for pumping up through the tubing string **16** of the assembly **10**. A remaining area surrounding the inner tube **30** to span between the inner tube and the surrounding outer tube **18** typically comprises an inner annular passage **34** as described in further detail below. The inner tube **30** also typically extends a height of the multistage separator section **22**.
[0018] An overall height of the multistage separator section **22** locating the inner tubes therein is further subdivided into a plurality of separator stages **36** mounted one above the other in series to form a column. A separate portion of fluid is communicated from the outer annular region **20** into the inner passage of the inner tube **30** at each separator stage **36** in which all of the fluid entering at each stage enters the inner tube **30** at a common elevation which is different from the elevation of the other separator stages.
[0019] The assembly **10** of FIGS. **1-3** further includes a diversion apparatus which defines a

diversion path **38** therethrough that is isolated from the inner passage of the inner tube **30** and which communicates from a bottom end of the diversion apparatus **30** in open communication with the intake opening of the intake section **24** to a top end in open communication through the outer tube **18** to the outer annular region **20**.

[0020] The diversion apparatus functions as a diversion separator section **40** in addition to the multistage separator section **22**. In the embodiment according to FIGS. **1** and **2**, the diversion separator section **40** extends alongside the multistage separator section **22** such that the two sections longitudinally overlap one another; however, in the embodiment of FIG. **3**, the diversion separator section **40** is located fully below the multistage separator section **22** such that the sections **40** and **22** are longitudinally in series with one another.

[0021] In either embodiment, the multistage separator section **22** locates one or more limited entry ports **42** at each separator stage that collectively define a combined cross-sectional flow area. The limited entry ports **42** each comprise a restricted orifice of fixed dimension in the communication path between the outer annular region and the inner passage of the inner tube **30** to provide restricted communication of fluids from the outer annular region to the inner passage of the inner tube. For clarity, the phrase "limited entry" means that the ports are sized such that the flow rate of liquid material from the inner annular region **34** of each stage **36** into the inner tube **30** is restricted by the size of the port **42**. This is to contrast limited entry ports with ports typically seen in conventional separators, in which the inner tube **30** or similar structure has no restriction of fluid.

[0022] Typically the cross-sectional flow area of all limited entry ports at any one level or any one separator stage is smaller than the combined cross-sectional flow area of all limited entry ports **42** of any one level or separator stage therebelow. The sizing may be arranged such that despite the pressure differential existing between each separator stage at different elevations relative to one another, a similar volume of fluid is intended to be drawn from each separator stage at each corresponding elevation. In this manner fluids are drawn from the outer annular region into the inner tube at a substantially even distribution across a plurality of different elevations along the limited-entry multiple chamber separator **22**.

[0023] Alternatively, each port could also be the same size (e.g., 1/16″, ⅛″, etc.) or there could be several of the same size with one or more of larger size below them (e.g., (3)x 1/16″ and (2)x⅛″ below). Accordingly, the areas of the limited entry ports **42** are not necessarily all to be one size, with increasing size from top to bottom preferred.

[0024] In both embodiments, the bottom end of the inner tube **30** includes a bottom entry port **44** that also receives fluid from the outer annular region into the inner passage of the inner tube, but at a location which is below all of the separator stages of the multistage separator section **22**, such that the bottom entry port **44** is spaced below all of the limited entry ports **42**. The bottom entry port **44** may locate a pressure responsive variable valve **46** therein in which the valve is operable through a range of cross-sectional flow areas from a fully open position of the valve defining a maximum cross-sectional flow area through the bottom entry port to a fully closed position in which flow through the bottom entry port is shut off.

[0025] When provided, the variable valve **46** operates in response to pressure differential across the valve between the outer annular region and the bottom of the inner passage of the inner tube. When the pressure differential is below a minimum threshold, the variable valve **46** remains closed such that all fluid drawn into the inner passage of the inner tube is only drawn through the limited entry ports **42** of the multistage separator section. The fluid in the outer annular region has already been drawn through the diverter separator section **40** in this instance. When the pressure differential exceeds the prescribed minimum threshold setting of the valve, the valve begins to open and continues to open by larger amounts as the pressure differential increases until the valve reaches the fully open position when the pressure differential exceeds a maximum threshold. The cross-sectional flow area of the bottom entry port in the fully open position of the variable valve is equal to or greater than the cross-sectional flow area of the limited entry ports of any one separator stage, and more preferably is greater than the cross-sectional flow area of the limited entry ports of all separator stages combined. In this instance, if all of the limited entry ports **42** became plugged, the assembly can maintain overall flow rates therethrough by fully opening the variable valve. When the variable valve is open, flow can be redirected directly from the diversion separator section **40** into the inner passage of the inner tube, thereby bypassing the plugged limited entry ports of the multistage separator section **22**.

[0026] Alternatively, when there is no variable valve **46** in the bottom entry port **44**, the bottom entry port may be partially restricted; however, the flow area of the bottom entry port preferably remains equal to or greater than a combined flow area of all limited entry ports **42**.

[0027] As described herein, in each instance of the first three embodiments when a variable valve **46** is provided, the assembly is operable in a first mode providing diversion separation from the bottom intake opening to the outer annular region through the diversion separator section **40** followed by an even distribution of fluid from the outer annular region to the inner tube across the plurality of different separator stages **36** at different elevations. As the limited entry ports **42** become plugged, the assembly assumes a second mode of operation in which the variable valve **46** opens proportionally to the amount of reduced flow from the limited entry ports to compensate for the reduced flow through the multistage separator section and maintain overall flow through the assembly **10**.

[0028] Turning now more particularly to the first and second embodiments of FIGS. **1** and **2**, the multistage separator section in this instance longitudinally overlaps the diversion separator section **40**. Accordingly, the intake section **24** in this instance is formed directly at the bottom end of the outer tube **18** such that the intake opening comprises the open bottom end of the outer tube, and the diversion flow path through the diverter separator section extends upwardly through the inner annular region of the outer tube, alongside the inner tube **30**. The top end of a diversion passage communicates through the outer tube **18** such that the top end of the diversion passage is in open communication with the outer annular region above the separator stages of the multistage separator section **22**. Furthermore, the bottom end of the inner tube **30** communicates through the outer tube **18** at the bottom end of the outer tube immediately above the isolation plug **28** at a location below the separator stages **36** such that the bottom end of the inner passage of the inner tube draws fluid from the bottom of the outer annular region as regulated by the variable valve **46** within the bottom entry port **44**.

[0029] Within the embodiment of FIG. **1** in particular, the orifices defining the limited entry ports **42** are located within the outer tube **18** such that each limited entry port communicates through a respective tubular channel **48** from the limited entry port in the outer tube to the inner passage of the inner tube **30** at the respective elevation of the separator stage **36**. The remainder of the inner annular region surrounding the inner tube **30** in this embodiment defines part of the diversion passage communicating from the open bottom end of the outer tube **18** to the discharge outlets at the top end of the outer tube that communicate the fluid from the intake opening to the outer annular region.

[0030] Within the embodiment of FIG. **2**, a diversion tube **50** is provided within the interior of the outer tube **18** to run alongside the inner tube **30**. The diversion tube **50** has a bottom end in open communication with the intake opening at the bottom of the assembly to define a portion of the diversion passage **43** of the diversion separator section which extends upwardly along the length of the outer tube **18** to a top end communicating through the outer tube into the outer annular region. In this embodiment, each of the separator stages **36** comprises a segmented portion of the inner annular region about the inner tube **30** and the diversion tube **50** which is enclosed at longitudinally opposing ends by respective isolation members **52**. Each separator stage **36** thus comprises a respective separator chamber **54** which is isolated longitudinally between a pair of the isolation members **52**. At each separator stage **36**, an intake slot **56** comprising one or more openings communicates through the outer tube **18** so that fluids from the outer annular region may enter the respective separator chamber **54**.

[0031] The limited entry ports **42** associated with the respective separator stage are in turn located within the inner tube **30** adjacent the bottom end of the chamber **54** of that separator stage so that fluid from the separator chamber communicates through the limited entry ports into the inner passage of the inner tube. The intake slots **56** of each stage are arranged to be less restrictive than the limited entry ports of the same stage such that fluid may readily enter through the intake slots into the separator chambers **54**, however, gas can also escape from the separator chambers **54** back into the outer annular region through the intake slots **56**. Both the diversion tube **50** and the inner tube **30** communicate through each isolation member **52** to maintain isolation between the interiors of the tubes and the surrounding separator chambers **54** with the exception of the communication of fluid from the separator chambers into the interior passage of the inner tube through the limited entry ports **42** as described herein.

[0032] Turning now to the third embodiment shown in FIG. **3**, the multistage separator section **22** in this instance is located fully above the diversion separator section **40** such that the sections are longitudinally in series with one another.

[0033] The multistage separator section **22** in FIG. **3** is similar to the previous embodiment of FIG. **2** in that the inner annular region between the inner tube **30** and the outer tube **18** locates a plurality of isolation members **52** defining upper and lower boundaries of respective separator stages that each comprise a separator chamber **54**. Also as in the previous embodiment, in the embodiment of FIG. **3**, intake slots **56** communicate through the outer tube **18** at the top of each chamber **54** and the limited entry ports **42** are located in the inner tube **30** at the bottom of the respective chamber **54**.

[0034] The diversion separator section **40** in this instance includes its own outer tubular member **60** connected in series with the outer tube **18** thereabove and the intake section **24** therebelow. The intake section in this instance comprises its own tubular member. An upper isolation member **62** is mounted within the outer tubular member to span the interior of the outer tube adjacent the top end thereof. The remaining interior of the outer tubular member **60** above the upper isolation member **62** comprises an upper region **64** in open communication with the bottom entry port **44** at the bottom of the inner tube located thereabove. The variable valve of **46** in this instance regulates flow from the upper region **64** of the diversion separator section **40** to the bottom end of the inner passage of the inner tube **30** thereabove.

[0035] An inner tubular member **66** is mounted within the outer tubular member **60** to span substantially the full length of the outer tubular member. The top end of the inner tubular member **66** communicates through the upper isolation member such that an inner passage extending longitudinally inside the inner tubular member **66** is in open communication at the top end thereof with the upper region **64**. An intermediate annular region **68** is defined between the inner tubular member and the outer tubular member to extend longitudinally along the length of the diversion separator section **40**. Discharge ports **70** are provided in communication through the outer tubular member **60** adjacent the top end thereof immediately below the upper isolation member **62** such that the intermediate annular region **68** is in communication at the top end thereof with the surrounding outer annular region **20** and the outer annular region between the outer tubular member **60** and the surrounding casing. The outer annular region **20** surrounding the outer tubular member **60** is in open communication with the outer annular region surrounding the outer tube **18** thereabove. The bottom end of the intermediate annular region **68** within the outer tubular member **60** is in open communication with the intake opening of the intake section **24** therebelow. A plurality of radial tubes **74** are mounted in communication from the bottom end of the inner tubular member **66** to the outer annular region **20** adjacent the bottom end of the outer tubular member **60** at a location immediately above the isolation plug **28**. The bottom end of the inner tubular member **66** is otherwise closed such that any fluid entering the inner tubular member **66** can only enter from the bottom end of the outer annular region **20** through the radial tubes **74** providing an open communication path therebetween.

[0036] The diversion flow path **38** in this instance extends from the intake section **24** up through the intermediate annular region **68** alongside the inner tubular member **66** prior to being diverted into the outer annular region **20** through the discharge ports **70** where the fluid can undergo some degree of separation where gasses can rise up through the outer annular region of the assembly and other fluids can fall down through the outer annular region **20** to enter into the inner passage of the inner tubular member via the radial tubes **74**.

[0037] In the first mode of operation when the variable valve is closed, fluid enters the inner passage of the inner tube **30** of the multistage separator section thereabove only by being drawn through the limited entry ports **42** from the outer annular region, through the separator chambers and into the inner passage of the inner tube **30**. In the event of any reduced flow through the limited entry ports **42**, the resulting pressure differential causes the variable valve **46** to assume a second mode of operation in which the valve at least partially opens so that some fluid can be drawn from the bottom end of the outer annular region **20** through the inner tubular member **66** of the diversion apparatus.

[0038] With reference now to FIG. **4**, a fourth embodiment will now be described in further detail. In this instance, the assembly **10** comprises a multistage separator section **22**, substantially identical in configuration to the multistage separator section **22** of FIG. **3**, but with more individual separator stages **36** included therein. In this instance, each separator stage **36** again comprises a chamber **54** within the inner annular passage **34** between the outer tube **18** and the inner tube **30** and which is isolated at longitudinally opposing ends by respective isolation members **52**. Each chamber **54** receives fluid through an intake slot **56** and the outer tube **18** and allows fluid to be drawn from the chamber into the inner tube **30** through the one or more limited entry ports **42** associated with that stage in the same manner as the previous embodiment.

[0039] The assembly **10** of FIG. **4** differs from the previous embodiment in that the multistage separator section **22** in this instance is used separately from any diversion separator section **40** so that no isolation plug **28** or intake section **24** is required. Instead, the outer annular region **20** between the casing **12** and the outer tube **18** receives fluid from the wellbore therebelow by being in open communication at the bottom end thereof to the wellbore therebelow. The bottom end of the outer tube **12** is enclosed by a lowermost one of the isolation members **52**. In the illustrated embodiment, the bottom end of the inner tube **30** is also enclosed at the bottom end thereof so that fluid only enters the inner tube through the limited entry ports within the various stages **36** of the multistage separator **22**.

[0040] In a variation of the embodiment of FIG. **4**, the bottom end of the inner tube **30** may communicate with the wellbore therebelow through a bottom entry port **44** at the bottom end thereof, in which a variable valve **46** (such as that of FIG. **3**) is provided within the bottom entry port **44** to regulate flow therethrough. Similarly to the description of the variable valve **46** in the previous embodiments, the valve is normally closed such that all fluid must be drawn through the limited entry ports across the various stages **36**; however, as the limited entry ports become plugged, the variable valve **46** may open responsive to the variation in pressure to allow some flow to be diverted through the bottom entry port.

[0041] As a further alternative, the inner tube **30** may be open at its bottom such that the bottom stage does not have a limited-entry port.

[0042] It should be appreciated that in FIGS. **1-4**, the solid arrows represent liquid flow, while the outlined arrows represent gas flow. In every embodiment of the invention, the goal is to allow gas to flow up the annulus of the wellbore while liquid settles into the separator and is removed using the inner tube **30**, which is in communication with a pump.

[0043] With reference to FIG. **5**, the outer tube **18** of a single separator stage **36** is shown. The outer tube **18**, as shown in preceding figures, has intake slots **56** disposed for fluid flow between each separation stage **36** and the outer annular region **20**. As shown in FIG. **5**, these openings are small relative to the length of the separation stage **36**. For example, the intake slots **56** may be approximately one-sixth the length of the stage.

[0044] Such openings are satisfactory for separation operations. That is, they may properly allow wellbore fluid to transit into the “quiet-space” of the separator stage **36** and allow gas, once separated, to transit out of the stage and back into the annular region **20**. That said, the placement of any apparatus, such as the multistage separator **22** (FIGS. **1-4**), within the casing **12**, restricts the cross-sectional area of the annular region **20**. As a result, while flowrate remains approximately constant, velocity will increase. Such increased velocity increases the risk that liquid will flow past all of the intake slots **56** of the separator and remain in the annulus, reducing the effectiveness of the separation.

[0045] With reference now to FIGS. **6** and **7**, the outer tube **18** comprises an annular velocity disruptor (“AVD”) apparatus **100**. The AVD apparatus **100** comprises multiple large intake openings **102**. In FIG. **6**, there are four such openings **102** shown. In FIG. **7**, there are five such openings **102** shown, though each set of openings **102** are phased at ninety degrees, such that three rows of four openings **102** are distributed about the perimeter of the apparatus **100**. It should be understood that these openings extend about the perimeter of the apparatus **100** and are roughly evenly distributed (including on the back of the AVD apparatus **100** shown). It may be advantageous, where structural integrity allows, to stagger the distribution of the openings **102**, as shown in FIG. **7**. When thick materials are used, all the openings **102** may be cut along the same axis.

[0046] By increasing the size of the openings **102**, a significant portion of the height of each stage **36** will include an opening to the “quiet-space” of each stage **36**. For example, it may be preferable to include a minimum of sixteen to twenty inches of axial opening and as much as thirty inches of axial opening on each stage **36** to create annular velocity disruption. This effectively increases the cross-sectional area of the annular region **20** by allowing for very easy transmission of wellbore fluids across an outer wall **104** of the AVD apparatus **100**. The length of the opening **102** may be dictated by modeled downhole well velocities, fluid properties (such as the foaminess of the material) etc. This is in contrast to common separators which have very narrow slots that are between an eighth of an inch and four inches wide, and only six inches tall. Such small slots are only thought to serve as a point of intake to the inside of a separator body.

[0047] Optimal dimensions are dependent upon the length of each stage **36**. For example, for a 48″ stage, the openings **102** may be 8″-16″, more or less, with a 2″-3″ width. For a 24″ stage, the openings **102** may be 8″-12″ in length, with a 2″-3″ width. Dimensions are provided as examples only and are not restrictive, and wider openings **102** are certainly possible. It is advantageous for the openings **102** to be wide enough and long enough, relative to the stage **36** length, to meaningfully disrupt or stall the flow of material in the annular space **20**. It may be advantageous for each opening **102** to have a width no less than one tenth of its height.

[0048] The AVD apparatus **100** may be used on one or more of the stages **36** of the multistage separation apparatus **22**. By increasing the cross-sectional area of the annular section **20**, fluid flow is slowed or completely stalled, creating an area across the AVD apparatus **100** that is liquid loaded with a higher concentration of fluid. This enhances separation at each stage and decreases the likelihood for liquid to flow by the entire separator **22**.

[0049] Since various modifications can be made in my invention as herein above described, and many apparently widely different embodiments of same made, it is intended that all matter contained in the accompanying specification shall be interpreted as illustrative only and not in a limiting sense.

[0050] Changes may be made in the construction, operation and arrangement of the various parts, elements, steps and procedures described herein without departing from the spirit and scope of the invention as described in the following claims.

Claims

1. A method of separating a multi-phase fluid into a liquid component and a gaseous component, the method comprising: receiving the multi-phase fluid into a first annular region defined between a casing and an outer tube; transmitting at least a portion of the multi-phase fluid from the first annular region into a plurality of separator stages defined within a second annular region between the outer tube and an inner tube, each separator stage bounded by a first isolation member and a second isolation member; within each separator stage, separating the multi-phase fluid into a gaseous component and a liquid component; directing the gaseous component from each separator stage into the first annular region; directing the liquid component from each separator stage into the inner tube through a corresponding limited entry port, wherein the limited entry ports of the plurality of separator stages are arranged in order of increasing cross-sectional area from the topmost stage to the bottom-most stage.
2. The method of claim 1 further comprising disrupting a velocity of the multi-phase fluid using an annular velocity disruptor apparatus.
3. The method of claim 2 in which the annular velocity disruption apparatus defines a slot having a width of at least two inches.
4. The method of claim 3 in which the slot has an axial length of at least eight inches.
5. The method of claim 1 further comprising allowing fluid to flow from the first annular region into the inner tube through a bottom entry port located below the plurality of separator stages.
6. The method of claim 5 further comprising regulating flow through the bottom entry port with a pressure-responsive variable valve.
7. The method of claim 6 in which the pressure-responsive variable valve adjusts flow into the bottom entry port in response to pressure differential between the first annular region and the inner tube.
8. The method of claim 6 in which the pressure-responsive variable valve opens proportionally in response to reduced flow through one or more of the plurality of limited entry ports.
9. The method of claim 1, wherein each of the plurality of separator stages is configured to intake fluid at a different elevation than an adjacent one of the plurality of separator stages.
10. The method of claim 1 in which each of the limited entry ports is a different size.
11. The method of claim 1 further comprising pumping the liquid component to a surface location through the inner tube.
12. The method of claim 1 further comprising plugging the first annular region below the bottom-most of the plurality of separator stages.
13. The method of claim 1 in which at least one of the plurality of separator stages has an axial length of forty-eight inches or more.
14. The method of claim 1 further comprising: producing the liquid component at a surface location from the inner tube; and producing the gaseous component at a surface location from within the casing.
15. The method of claim 1 in which each of the plurality of separator stages defines an intake slot within the outer tube, wherein the intake slot is approximately one-sixth the axial length of its associated separator stage.
16. The method of claim 1 in which no multi-stage fluid may directly enter a separator stage from the second annular region of an adjacent separator stage.
17. A method, comprising: diverting portions of a multi-stage fluid travelling upward through an annular space between a casing and a separator body into the separator body through spaced-apart intake openings within a set of vertically stacked and isolated separator stages arranged along the body; causing the multi-stage fluid to slow, settle, and separate into a gaseous portion and a liquid portion within each of the isolated separator stages, such that the gaseous portion escapes into the

annular space and the liquid portion remains; and pumping the liquid portion from each of the isolated separator stages through a corresponding port, into an internal tube that directs the liquid portion upward toward a surface, wherein none of the ports have a smaller cross-sectional area than the port within the top-most of the stages, and none of the ports has a larger cross-sectional area than the port within the bottom-most of the stages, and at least two of the ports have different cross-sectional areas.

18. The method of claim 17 further comprising: regulating flow through the bottom entry port with a pressure-responsive variable valve.